

serial, SPI (serial peripheral interface), I²C (inter IC bus), DMX (communication protocol for lighting and effects), Ethernet, Bluetooth, Wi-Fi

For more examples, you may visit www.arduino.cc and www.freeduino.org

Shields and extensibility

Shields offer a lot of flexibility and scalability in building applications using Arduino and Freeduino. These are basically the boards which plug on top of the Arduino and Freeduino boards. The package-on-package concept imparts modularity to applications development. Different types of shields are available. You can also design your own shield to suit your needs.

Commonly available shields are:

1. Ethernet shield (enables communication with other computers over local-area network)
2. Motor shield (allows control of servo and stepper motors)—ideal for robotics
3. XBee shield (enables wireless connectivity using ZigBee 802.15.4 protocol)
4. Prototyping shield (similar to breadboarding)
5. GPS shield (enables global positioning system based applications)
6. DMX shield (allows control of lighting, etc)

Arduino/Freeduino board details

Generally, an Arduino/Freeduino board consists of a DC jack to power the board. Powering the board is simple: use a 9V DC adaptor with 1 amp or a commonly available 9V DC battery. The board has 14 digital general-purpose input/output (I/O) pins and six analogue pins. It is generally equipped with either a serial port or a USB port for PC connectivity.

Some of the digital pins serve dual purpose: They can either act as digital general-purpose I/O or be used for serial connectivity, PWM outputs, I²C and SPI outputs. Ar-



Arduino LED blinking example

duino or Freeduino boards using ATmega 8 are equipped with three pulsewidth-modulated (PWM) outputs, whereas boards using ATmega 168 are equipped with six PWM outputs.

Why and who should use one? Like I said, a lot of time and effort was spent getting these boards ready for use and they work on the principle "Build it and share it." They are open to all, less expensive, very easy to program and lots of applications are possible with a single board.

Arduino and Freeduino are ideal for research and engineering projects, hobbyist and DIY (do-it-yourself) projects, and college projects.

Types available. Arduino and Freeduino are available in two types depending on the memory and connectivity requirements:

1. Memory: 8 kB using ATmega8 or 16 kB using ATmega168
2. Connectivity: Serial port or USB

How to start building applications?

1. Get the Freeduino board. To purchase one, visit 'www.freeduino.org' or 'www.bhasha.co.cc'. You may also make your own board using the

CAD files available on the website

2. Download and install the software

3. Write your test program and upload it to the board. There are also sample programs available in the software

The Arduino software works on Windows, Linux and Mac OS.

Where can you get Arduino/Freeduino in India?

Searching on the Internet, I came across website 'BHASHA E-STORE' by Bhasha Technologies. They sell the Arduino and Freeduino boards and kits in India,

and their boards are quite cheap. You may visit their website 'www.bhasha.co.cc' or e-mail to 'sales@bhasha.co.cc'.

I had recently purchased a kit from them which included Freeduino serial board, serial cable, 9V DC adaptor, and a software CD containing the free software, tutorials, datasheets and e-Book.

Sanguino: the next-generation board

The next generation of Freeduino board is Sanguino, which boasts a powerful ATmega 644/644P 40-pin IC. Visit 'www.sanguino.cc' for more details. The Sanguino board is fully open source and has a large number of input/output pins for most demanding needs. It features 32 general-purpose I/O pins (some are multipurpose), eight analogue pins, six PWM pins, 64kB flash memory, 4kB RAM and 2kB EEPROM. ●

The author is working as a technical architect at Mastek, a software exports company in India. He has over nine years of experience in designing and architecting complex business applications for clients worldwide, and is a keen hobbyist and automation enthusiast who likes to build prototype applications using Freeduino boards